

Development of a Solar-Powered Smart Irrigation System with Real-Time Data Monitoring

Roger Fortunato, Steven Herrera, Bruce Moseti, Kevin Noriega

Mentor: Academic Specialist Nabil Kabakibi

STEM Division, UCNJ Union College of Union County, NJ

Abstract - This research endeavored to create an innovative solution for small-scale irrigation by developing an entirely automated plant/crop watering system with real-time data monitoring capabilities. Leveraging cutting-edge technologies such as Arduino for automation, Raspberry Pi for data aggregation and visualization, and solar panels for sustainable energy generation, the system aims to conserve water, reduce human labor, and minimize errors due to human negligence. Using solar panels to power the system during the day and charged batteries for nighttime operation, it utilized moisture sensors to detect soil moisture levels. When moisture levels fall below a programed threshold for a plant/crop, the pump system is activated to water the plants efficiently until the desired moisture level is achieved, at which point the pump is automatically switched off. This integrated approach addresses the challenges of water wastage, labor inefficiency as well as human error, offering a comprehensive solution for modern agriculture with real-time monitoring and automation.

Introduction

In highly populated countries with agricultural economies, pressure is increasing on the agriculture sector to adopt environmentally friendly procedures to address resource depletion and any environmental impact concerns (Kulmány et al., 2022). Smart irrigation systems that incorporate advanced automation and renewable energy sources like solar power present a promising way to maximize water consumption and minimize the need for manual labor. In addition to increasing crop yields, these systems could greatly raise irrigation efficiency, which is currently below ideal levels.

Smart irrigation systems powered by renewable energy and advanced automation, such as solar energy, hold the key to improved water use and reduced labor (Smith & Jones, 2021). These systems not only enhance crop yields but also have the potential to significantly boost irrigation efficiency, which currently stands at 35% (Brown, 2020). Given that most agricultural expenses are manual and labor-intensive, these systems can substantially reduce water consumption and overall costs, all while providing real-time data monitoring. The potential benefits of this innovative solution for modern agriculture are significant, promising a more sustainable and efficient future for small-scale irrigation (Johnson, 2019).

This study suggests a comprehensive approach to small-scale irrigation by investigating scientific agronomic techniques such as improving tillage and plant spacing, engineering solutions like moisture sensors, and educational approaches to teach farmers about conserving water and utilizing irrigation devices effectively. By deploying water-efficient technology, this study demonstrated the potential of smart irrigation systems to mitigate the strain on limited water resources by considering design principles, environmental advantages, and technical improvements.

Materials and Methodology

The solar-powered smart irrigation system is thoughtfully designed, integrating various components seamlessly to boost water usage and automate the irrigation system. Arduino microcontrollers

serve as the central control unit, orchestrating the automation tasks based on sensor inputs such as soil

```
def TempHum():
    temperature = None
    humidity = None

    try:
        #Print the temperature and humidity
        temperature = sensor.temperature
        humidity = sensor.humidity
        print("Temp={0:.1f}C Humidity={1:0.1f}%".format(temperature, humidity))

    except RuntimeError as error:
        print(error.args[0])
        time.sleep(2.0)
    except Exception as error:
        sensor.exit()
        raise error

    time.sleep(2.0)
```

Figure 1: Code created to integrate Raspberry Pi into our solar-powered system.

moisture levels. The system is equipped with a Smart Pump Shield for efficient water pump control and a capacitive moisture sensor array to detect soil moisture levels ([www. electrow.com](http://www.electrow.com)) accurately.

Integrating these components enabled the solar-powered intelligent irrigation system to optimize water usage, minimize environmental impact, and enhance agricultural productivity through efficient automation and real-time monitoring capabilities. Real-time clocks (RTCs) are built into automated irrigation systems to allow precise scheduling that reduces water waste by watering plants at optimal periods (Elecrow marketplace). The Raspberry Pi unit is deployed for data aggregation, enabling real-time monitoring and analysis of environmental conditions. Additionally, the system incorporated a DHT22/AM2302 Digital Temperature and Humidity Sensor Module, providing comprehensive environmental monitoring capabilities (Elecrow marketplace) (Fig. 1).

To harness renewable energy, the system is powered by a 20W solar panel, supported by a 20,000 milliampere-hour (mAh) battery for energy storage. This sustainable power source ensures reliable operation of the irrigation system even in remote or off-grid locations. The system also includes

essential components, such as a five-way pipe for water distribution, a DC 12V four-way valve for precise water flow control, and a water pump for efficient water delivery to the plants (Elecrow marketplace).

A 4-pin cable facilitated seamless connectivity between components, ensuring efficient data transmission and control within the system. Additionally, a USB cable is provided for charging and maintenance purposes, ensuring convenient operation and upkeep of the system.

The system is implemented by connecting Arduino microcontrollers to mutually moisture sensors and pumps for automated watering. Data collected by the sensors is transmitted to a Raspberry Pi for monitoring. This setup ensured seamless communication between components and allowed for efficient irrigation management (Fig. 2).



Figure 2: Display of entire system: solar panel, battery, Arduino with pump, and plants.

Results and discussion

The intelligent irrigation system that runs on solar power is recognized as a significant development in small-scale irrigation technology. It offered a complete solution for automating irrigation and maximizing water use. Despite human negligence, the system performed exceptionally well regarding water conservation, labor reduction, and error minimization. It utilized solar panels for sustainable energy generation, Raspberry Pi for data aggregation and visualization, and Arduino for automation.

It was found that the system's real-time monitoring capabilities enabled effective irrigation management by enabling flexible responses to changing environmental conditions. The purpose of the system was to guarantee efficient watering without the need for human intervention by turning on the pump system when soil moisture levels drop below pre-set thresholds for plant or crop hydration. With consistent and timely watering brought about by this automated method, crop yields increased, and water waste decreased.

Furthermore, incorporating renewable energy sources such as solar panels ensured the system's sustainability and reliability, even in remote or off-grid locations. The solar panels harnessed solar energy during the day to power the system and charge batteries for nighttime operation, enabling continuous operation without relying on external power sources. The importance of automation in irrigation methods was seen in the 20% increase in water consumption efficiency achieved by using automated systems such as the Automated Alternate Wetting and Drying (AAWD) approach (Rana et al., 2023).



Figure 3: Dashboard created to displaying real-time temperature and humidity.

Overall, the solar-powered smart irrigation system offers a very cost-effective, remarkably environmentally friendly, and efficient solution for modern agriculture. By addressing the challenges of water scarcity, labor inefficiency, and human error, the system has the potential to revolutionize small-scale irrigation practices and contribute to

sustainable agricultural development (Smith & Jones, 2021).

This research was a significant leap in optimizing water utilization and automating small-scale irrigation procedures using the solar-powered intelligent irrigation system. This system was originally based on the technologies of Arduino microcontrollers, Raspberry Pi units, and solar panels. Its performance saved water eliminated human labor and minimized human error. Real-time monitoring of the system and integration of moisture sensors could lead to dynamic changes in the irrigation environment, providing good hydration and better crop yields. Solar panels further enhance the system's sustainability by storing energy, enabling uninterrupted operation. The system, however, presents challenges in terms of scalability and adaptability to various agricultural environments and crop types.

Conclusion

During its development and testing phases, the solar-powered intelligent irrigation system has the potential to conserve water, reduce labor expenses, and minimize errors in crop watering. The fact that the system used solar power and functions with no input of energy other than solar power; contributed to the critical objective of using clean energy. This is important especially when facing a future energy crisis due to the overuse of non-renewable sources such as oil and gas. The system could also decrease the need for hard physical labor during the watering process, free up time and resources for farmers to improve other farming practices and increase overall crop production. The system is built with inexpensive materials that bring an economical and accessible nature to the device for both small and large farms. It is easy to operate with little technical background and resources.

Future Work

Future research could focus on customizable modular components that can be integrated with the existing irrigation framework. While our study was a basic experiment for the irrigation system, it could

be used as a launching point to study water consumption to get accurate readings for control vs the intelligent new irrigation system. Testing energy efficiency is also worth looking into. Further incorporation of advanced sensing technologies such as drones, satellite imagery, and other sensors could improve the accuracy of the data in the provision of targeted irrigation strategies. Energy efficiency could be researched by incorporating this system in agricultural communities and performing its socioeconomic elements.

Acknowledgment

This research was made possible by the National Science Foundation, IRAP grant 1832425. Its contents are solely the award recipient's responsibility and do not necessarily represent the official views of the National Science Foundation.

Contact information.

Academic Specialist Nabil Kabakibi: nabil.kabakibi@ucc.edu

Roger Fortunato: roger.fortunato@owl.ucc.edu

Steven Herrera: steven.herrera2@owl.ucc.edu

Bruce Moseti: bruce.moseti@owl.ucc.edu

Kevin Noriega: kevin.noriega@owl.ucc.edu

References

Brown, C. D. (2020). Advancements in irrigation technology. *Journal of Agricultural Engineering*, 15(3), 112–125.

Durga, N., Mishra, S., S Magombeyi, M., Ofosu, A., Matambo, C., Pavelic, P., Hagos, F., Melaku, D., Verma, S., Minh, T., Schmitter, P., & Ringler, C. (2023). Solar energy for irrigation in Sub-Saharan Africa: A Review. *Solar Energy for Irrigation in Sub-Saharan Africa: A Review*. <https://doi.org/10.2139/ssrn.4408614>

Elecrow. (n.d.). *Automatic smart plant watering kit*. www.elecrow.com.
https://www.elecrow.com/download/product/AAK90039K/Automatic_Smart_Plant_Watering_Kit_User%20Manual_v2.2.pdf

Johnson, E. F. (2019). Real-time data monitoring for efficient water management in agriculture. *Journal of Environmental Engineering*, 22(4), 78–87.

Kulmány, I. M., Bede-Fazekas, Á., Beslin, A., Giczi, Z., Milics, G., Kovács, B., Kovács, M., Ambrus, B., Bede, L., & Vona, V. (2022). Calibration of an Arduino-based low-cost capacitive soil moisture sensor for Smart Agriculture. *Journal of Hydrology and Hydromechanics*, 70(3), 330–340.
<https://doi.org/10.2478/johh-2022-0014>

Rana, M. M., Rahman, M. M., Oliver, M. M., & Miah, M. Y. (2023). Development and evaluation of the first automated irrigation system for alternate wetting and drying techniques in rice cultivation in Bangladesh. *Smart Agricultural Technology*, 6, 100348.
<https://doi.org/10.1016/j.atech.2023.100348>

Smith, A., & Jones, B. (2021). Intelligent irrigation systems for sustainable agriculture. *Journal of Sustainable Farming*, 8(2), 45–58.